COMP4620/8620 – Advanced Topics in AI: Intelligent Robotics

Semester-2 2026 – Assignment 1
Convenor & Lecturer: Hanna Kurniawati
Due date: Wednesday, 26 August 2026 23.59 Canberra time
Grace period: 12 hours after the due date (ie, 24 Aug 2026 00.00-11.59am Canberra time)
Late submission: NOT permitted —submission after the end of the grace period will automatically receive a mark of 0.

You must read the following notes first before starting to work on the assignment.

1. This is an individual assignment with a maximum total mark of 100 points and worth 25% of your total mark in this class.
2. This assignment consists of three parts: Part A, Part B, and in-person Q&A. The maximum mark for Part A is 32 points. The maximum mark for Part B is 33 points. The in-person Q&A has two components: (1) Demo of your program (B3). (2) Oral Q & A related to the assignment and all materials covered in Week 1 to Week 3 (inclusive). The maximum mark for component (2) of the in-person Q&A is 34 points.
3. The use of Generative AI is allowed, but you must disclose it and provide all the prompts used. Failure to disclose constitute cheating.
4. Submission Instruction for Part A and Part B:
   (a) You must submit the following 2 files via Canvas before the due date.
      i. One .pdf file containing answers to all the non-programming part of the assignment
      ii. One .zip/.7z/.tar.gz file containing your source codes and test cases.
   (b) For the Programming component, you must:
      i. Write your program in Python. We provide a set of functionalities you can use to help in this assignment. These functionalities are written in Python.
      ii. Submit all of your source codes and test cases used in your report. You must place all files (and folders) under a single top-level folder, compress the top-level folder into a single file with one of the following extensions: .zip or .7z or .tar.gz, and submit the compressed file. If your program consists of multiple files in multiple folders, your compressed file should preserve the folder structure. Note that during the in-person Q&A, you must download this file from Canvas, unzip, and run them in the lab's computer to solve the test scenarios we provide. Hence, all the files you need to run your program, aside from the libraries already installed in the lab's computer, must be included in this file.
   (c) Late submission is NOT permitted. We do provide a 12 hours grace period. Within the grace period, you can still submit and resubmit your assignment. However, after the grace period ends, you will NOT be able to submit your assignment. Without a submission, you will receive a 0 mark for this assignment.
5. Information for the in-person Q&A:
   (a) The in-person Q&A will take place on Friday, 28 August 2026 in Skaidrite Darius Building Room N114. The Schedule is in https://bit.ly/comp4620A1QASched . If your name is not listed in the schedule, you must contact the teaching staff via Ed Forum of the class.
   (b) You are recommended to arrive at least 10 minutes before the start of your Q&A slot.
   (c) You must bring your ANU ID and show it to the staff marking your Q&A.
   (d) You will not be able to participate in the in-person Q&A if you do not submit Part A and Part B before the end of the grace period (see point 4 above).

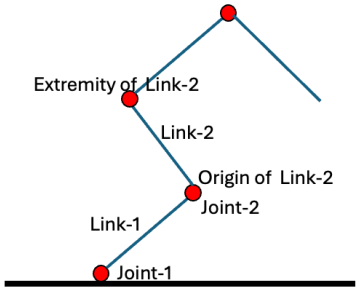
PART A [33 pts]

1. [10 pts] For the following questions, consider a 3D sphere robot with radius k ($k \in \mathbb{R}^+$) that can only translate, operating on a planar surface populated by polyhedral obstacles.

   (a) [5 pts] Can we use the visibility graph method to find a collision-free path between a given initial and goal configurations? If your answer is yes, please provide the Pseudo code. Otherwise, please explain why we cannot use the method.

   (b) [5 pts] Can we use uniform grid discretization to find a collision-free path between a given initial and goal configurations? If your answer is yes, please provide the Pseudo code. Otherwise, please explain why we cannot use the method.

2. [23 pts] Consider a planar kinematic chain robot as illustrated in the figure on the right. It has a static base, K rotational joints and K links. Each joint is represented as a point. Each link is a straight line segment with length L. It has two end-points, called the origin and the extremity points. The position of the origin of the first link is fixed. The origin of the ith link for $i \in [2, K]$ coincides with the extremity of the $(i-1)^{th}$ link at a joint point. The robot operates in a 2D workspace populated by a set of obstacles Obs.



   A configuration of the above robot can be represented by $q = (\theta_1, \theta_2, \cdots, \theta_K)$, where $\theta_i \in [0, 2\pi)$ is the joint angle that defines the angle (in radian) between the base and the first link for $i = 1$, and between the ith and $(i-1)^{th}$ link for $i = [2, K]$. The C-space of this robot can be represented as the space $\mathbb{R}^K$.

   In addition, the C-space distance between two configurations, $q = (\theta_1, \theta_2, \cdots, \theta_K)$ and $q' = (\theta'_1, \theta'_2, \cdots, \theta'_K)$, as: $d_C(q, q') = \max_{1 \le i \le K} |\theta_i - \theta'_i|$. This distance metric is often used in motion planning because it is faster to compute compare to the typical Euclidean distance.

   Please answer the following questions.

   (a) [15 pts] Given 2 configurations, $q = (\theta_1, \theta_2, \cdots, \theta_K)$ and $q' = (\theta'_1, \theta'_2, \cdots, \theta'_K)$, let us assume the robot moves from q to q' along a straight line segment $\overline{qq'}$ in the C-space. During such a movement, all points on the robot traces a path of length less than or equal to $\alpha \cdot d_C(q, q')$, where α is a constant that can be upper bounded in terms of the link length L and the number of links K. Find this upper bound of α and expressed them in L and K, and provide its derivation.

   (b) [8 pts] Now, recall that the workspace distance $d_W(q, Obs)$ between the configuration q and obstacles Obs in the workspace is defined as the distance between the closest pair of points on the robot placed at configuration q and Obs. Please find the radius τ of the neighbourhood $Neigh(q) = \{q' \mid d_C(q, q') \le \tau\}$ that will guarantee the robot can move from configuration q to q' (for any $q' \in Neigh(q)$) collision-free, following a straight line path $\overline{qq'}$. Please express τ in terms of the upper bound α from Q2(a) and $d_W(q, Obs)$.

PART B [33 pts]

Consider the problem of centralised motion planning for 5 drones navigating in an environment populated by obstacles. To simplify the problem, assume that each drone is represented by its bounding sphere and its movement is only translation in a 3D Euclidean workspace. The workspace is bounded and populated by obstacles. The geometry of the workspace along with the obstacles in it are known a priori. Each drone needs to move from a given initial to a given goal configurations in a collision-free manner. The drones should not collide with each other and should not fly outside of the workspace' boundaries. Your tasks are

to design a sampling-based method, implement it, conduct systematic experiments to identify the advantage and disadvantage of the method used, and report your findings. To guide you in these tasks, please answer the questions below:

1. [2 pts] Define the C-space we need to use for centralised motion planning of multiple drones, where the number of drones $K \in [1, 5]$. Centralised means that all drones in the multi-drones team are treated as a single robot by the motion planner.
2. [6 pts] Design a specific sampling-based motion planner to solve the centralised motion planning problem of multiple drones. Provide a pseudo-code of the motion planning method you will use to solve the problem.
3. [15 pts] Implement your pseudo-code, utilising primitive functionalities we provide (downloadable from `https://github.com/RDLLab/MultiDrone`). The mark for this question will come from the program's performance during the demo. Specifically, there will be 3 testing scenarios, differing in the number of obstacles. Your program has up to 20 seconds with no parallelisation to solve a given problem. Each solved problem is worth 5 points.
4. [6 pts] Empirically investigate the performance of the motion planning method you proposed as the environment complexity increases. To this end, you need to design workspace with increasing complexity (hint: consider our class discussion on the more theoretical side of sampling-based motion planning), and run systematic experiments on those environments. You must report the summary statistics (at least mean and 95% confidence interval) of the experiments and discuss how the increasing environment complexity affects the motion planner performance.
5. [4 pts] Discuss the scalability of the motion planning method you proposed when the number of drones increases.

Note that unlike Part A, the questions in this part are inter-related. For instance, your answer to question 3 must be an implementation of question 1 and 2, and your answers to questions 4–5 depend on your answers in question 1–3.

That's all folks